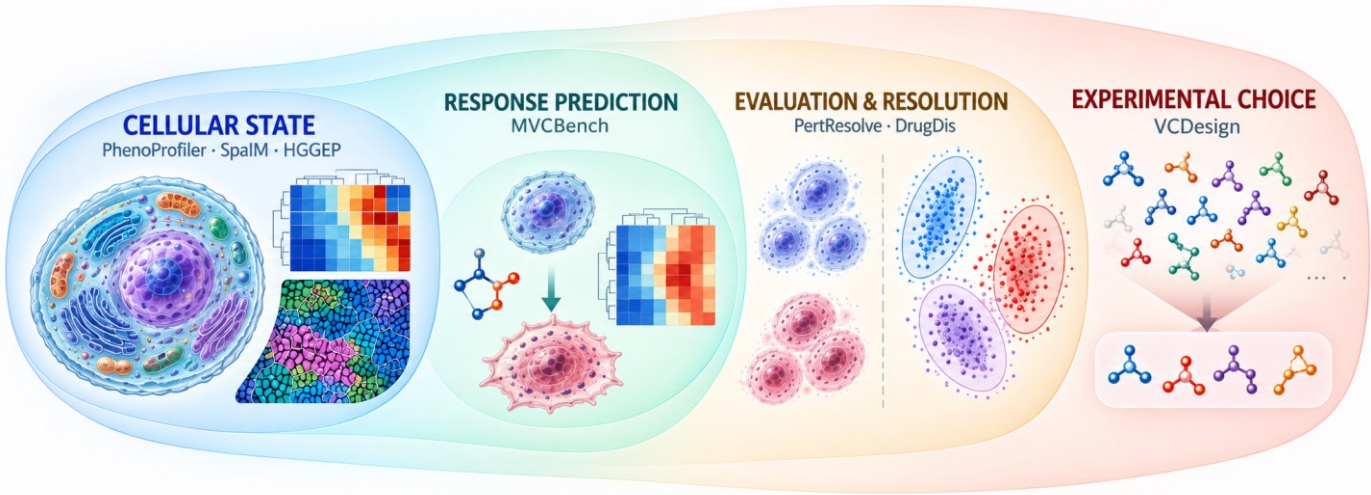


RESEARCH PROFILE

I develop decision-centric AI methods that connect multimodal biological measurements, virtual-cell modeling, and intervention design. My work asks what biological distinctions experiments can reliably resolve, what response structures models actually recover and generalize across cellular contexts, and which interventions are worth testing under limited experimental budgets.



EDUCATION & APPOINTMENTS

National University of Singapore	Jun 2026 – Jun 2027
Visiting Student Host: Prof. Yang Zhang	
University of Macau	Aug 2024 – Present
Ph.D. in Computer Science Advisor: Prof. Bob Zhang; Co-advisor: Prof. Qianqian Song	
Beijing University of Technology	Sep 2021 – Jul 2024
M.Eng. in Electronic Information Advisors: Prof. Yong Zhang and Prof. Baocai Yin	
Beijing Information Science & Technology University	Sep 2017 – Jul 2021
B.Eng. in Robotics Engineering Advisor: Prof. Hongbo Huang	

9	2	557	1,000+
First-author journal articles	First-author <i>Nature Communications</i>	citations · h-index 13 Google Scholar · Sep 2026	GitHub stars Across open-source projects

Snapshot: Sep 2026. First-author count includes all published journal articles; the list below is selected.

RESEARCH PROGRAM & SELECTED PUBLICATIONS

7 core works

CELLULAR STATE PhenoProfiler · SpaIM · HGGEF

How can heterogeneous biological measurements reveal cellular phenotypes and states?

- PhenoProfiler: Advancing Phenotypic Learning for Image-based Drug Discovery**
Bo Li, Bob Zhang, ..., Mengran Li, Yong Zhang, Qianqian Song · *Nature Communications* 17, 793 (2026)
Learns cellular representations from high-content microscopy for image-based compound profiling.
- SpaIM: Single-cell Spatial Transcriptomics Imputation via Style Transfer**
Bo Li, Ziyang Tang, ..., Baijian Yang, Jing Su, Qianqian Song · *Nature Communications* 16, 7861 (2025)
Connects single-cell and spatial transcriptomics through style-transfer-based expression imputation.
- Gene Expression Prediction from Histology Images via Hypergraph Neural Networks**
Bo Li, Yong Zhang, ..., Mengran Li, Guangyu Wang, Qianqian Song · *Briefings in Bioinformatics* 25(6), bbae500 (2024)
Uses hypergraph learning to capture higher-order structure for histology-to-gene-expression prediction.

RESPONSE PREDICTION MVCBench

What information transfers across perturbations, modalities, and cellular contexts?

4. **MVCBench: A Multimodal Benchmark for Drug-induced Virtual Cell Phenotypes**
Bo Li, Qing Wang, Shihang Wang, ..., Xiaojun Yao, Chuxia Deng, Qianqian Song · *bioRxiv* (2026)
Benchmarks multimodal perturbation prediction under compound, cell-line, plate, and dataset shifts.

EVALUATION & RESOLUTION PertResolve · DrugDis

When does predictive performance reflect reproducible, biologically useful response information?

5. **Measurement resolution constrains fine-grained perturbation prediction**
Bo Li, Chengyang Zhang, Mengran Li, ..., Yuhao Yi, Jiancheng Lv, Yang Zhang · *Manuscript* (2026)
Separates model limitations from measurement-resolution limits by testing which fine-grained perturbation distinctions experiments can reproducibly resolve.
6. **Disentangling general and context-specific effects in drug-response prediction**
Bo Li, Chengliang Liu, Yuzhong Peng, ..., Shenghui Huang, Chuxia Deng, Yang Zhang · *Manuscript* (2026)
Separates general drug and sample effects from drug-sample interactions, revealing when aggregate accuracy masks weak recovery of context-specific response structure.

EXPERIMENTAL CHOICE VCDesign

Which interventions are worth testing under a finite experimental budget?

7. **VCDesign: Candidate-Conditioned Inverse Modeling for Cellular Intervention Design**
Bo Li, Lin Wang, Bob Zhang, ..., Chengliang Liu, Zhiyuan Liu, Yang Zhang · *Manuscript* (2026)
Formulates finite-budget intervention ranking and evaluates selected candidates with independently measured held-out outcomes; transfers historical evidence to response-unseen candidates.

SELECTED OTHER FIRST-AUTHOR PUBLICATIONS

8. **Multi-scale Hypergraph-based Feature Alignment Network for Cell Localization**
Bo Li, Yong Zhang, Chengyang Zhang, Xinglin Piao, Yongli Hu, Baocai Yin · *Pattern Recognition* 149, 110260 (2024)
9. **Exponential Distance Transform Maps for Cell Localization**
Bo Li, Jie Chen, Hang Yi, ..., Yongquan Yang, Qikui Zhu, Hong Bu · *Eng. Appl. Artif. Intell.* 132, 107948 (2024)
10. **Lite-UNet: A Lightweight and Efficient Network for Cell Localization**
Bo Li, Yong Zhang, Yunhan Ren, Chengyang Zhang, Baocai Yin · *Eng. Appl. Artif. Intell.* 129, 107634 (2024)

Additional collaborative manuscript

11. **When do single-cell response distributions improve drug retrieval?**
Huihong Wang, Xiaoping Liu, Chen Chen, Qiuyan Sun[#], Bo Li[#], Bob Zhang[#] · *Manuscript* (2026)

OPEN SOURCE & RESEARCH SYSTEMS

Stars · 29 Sep 2026

1,000+ GitHub stars across open-source projects developed or co-developed; selected repositories:

Nature-Paper-Skills	Agent skills for scientific writing, revision, auditing and resubmission.	517 stars
Awesome Virtual Cell	Research landscape: 285 papers · 20 themes · interactive map.	311 stars
DrugClaw	Agentic RAG for drug research · 57 skills · 15 task categories.	116 stars

HONORS & AWARDS

National Scholarship	2022
Beijing Outstanding Graduate	2021, 2024
Top 100 Graduates, Beijing University of Technology	2024
Xiaomi Scholarship	2023
First-Class Academic Scholarship, Beijing University of Technology	2022, 2023
Ph.D. Scholarship, University of Macau	2024
First Prize, Science & Technology Innovation Scholarship, BISTU	2021

PATENTS

Co-inventor of three Chinese invention patent filings on cell localization and density-map generation:
CN115457546A · CN115810046A · CN115457547A

ACADEMIC SERVICE

Journal reviewer: Science Advances; IEEE TPAMI, TIP, TNNLS; Medical Image Analysis; Bioinformatics; Briefings in Bioinformatics; Pattern Recognition; BMC Biology; EAAI; Expert Systems with Applications.